

Two-Step Measurement Stories

Answer Key

Grade 2–3 measurement and data

Quick Answers

1. 75	4. 60	7. 165	10. 2300
2. 30	5. 830	8. 26	11. 350
3. 7.75	6. 1300	9. 2100	12. 8

Curriculum

Level: Grade 2–3 measurement and data

2.MD / 3.MD – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

2. If they got 55: took away only the minutes watched before dinner
9. If they got 2550: took out only one book instead of both
11. If they got 175: counted only the ride to school, not the ride home
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1. Swim lesson 45 minutes, then a 10 minute wait, then a 20 minute ride. How many minutes from the start of the lesson until Maya is home?

Solution: All three pieces of time happen one after another, so add them. *Step 1 — the lesson and the wait:* $45 + 10 = 55$ minutes. *Step 2 — add the ride home:* $55 + 20 = 75$ minutes.

75 minutes

2. A 95 minute movie; Ben watches 40 minutes, then 25 minutes. How many minutes are left?

Solution: *Step 1 — how much has he watched?* $40 + 25 = 65$ minutes. *Step 2 — take that away from the whole movie:* $95 - 65 = 30$ minutes. Check by adding back: $65 + 30 = 95$ ✓

30 minutes

3. Ella has \$20 and buys a \$8.75 book and a \$3.50 marker. How much is left, in dollars?

Solution: *Step 1 — add what she spent:* $\$8.75 + \$3.50 = \$12.25$. *Step 2 — take that away from what she had:* $\$20.00 - \$12.25 = \$7.75$. Line up the decimal points when you add and subtract money.

7.75

4. 3 quarters and 4 dimes, then 55 cents are spent. How many cents are left?

Solution: *Step 1 — count the money he starts with.* A quarter is 25 cents and a dime is 10 cents, so $3 \times 25 = 75$ cents and $4 \times 10 = 40$ cents, giving $75 + 40 = 115$ cents. *Step 2 — take away what he spends:* $115 - 55 = 60$ cents.

60 cents

5. Apples 250 g, pears 180 g, basket 400 g. What is the total mass?

Solution: Every mass is already in grams, so nothing has to be changed first. *Step 1 — the two bags of fruit:* $250 + 180 = 430$ g. *Step 2 — add the basket:* $430 + 400 = 830$ g.

830 g

6. A 2 L jug; 2 glasses of 350 mL are poured out. How many millilitres are left?

Solution: The jug is measured in litres and the glasses in millilitres, so change the litres first: $2 \text{ L} = 2 \times 1000 = 2000$ mL. *Step 1 — how much was poured out?* $2 \times 350 = 700$ mL. *Step 2 — take it away from the jug:* $2000 - 700 = 1300$ mL.

1300 mL

7. 25 minutes on each of 5 school days, plus 40 minutes on Saturday. How many minutes in the week?

Solution: *Step 1 — the school days:* $5 \times 25 = 125$ minutes. *Step 2 — add Saturday:* $125 + 40 = 165$ minutes.

165 minutes

8. 4 tickets at \$6 each, paid with a \$50 bill. How much change, in dollars?

Solution: *Step 1 — the cost of the tickets:* $4 \times \$6 = \24 . *Step 2 — subtract from the money handed over:* $\$50 - \$24 = \$26$.

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9. A 3 kg box; 2 books of 450 g each are removed. What is the mass now, in grams?

Solution: The box is in kilograms and the books are in grams, so change the kilograms first: $3 \text{ kg} = 3 \times 1000 = 3000$ g. *Step 1 — the mass of the books removed:* $2 \times 450 = 900$ g. *Step 2 — take it away from the box:* $3000 - 900 = 2100$ g.

2100 g

10. A 5 L cooler; 1200 mL are used in the morning and 1500 mL in the afternoon. How many millilitres are left?

Solution: Change the litres to millilitres first: $5 \text{ L} = 5 \times 1000 = 5000$ mL. *Step 1 — the water used all day:* $1200 + 1500 = 2700$ mL. *Step 2 — take it away from the full cooler:* $5000 - 2700 = 2300$ mL.

2300 mL

11. 2 bus rides a day on each of 5 school days, 35 minutes a ride. How many minutes in the week?

Solution: *Step 1 — how many rides altogether?* $2 \times 5 = 10$ rides. *Step 2 — how long are 10 rides?* $10 \times 35 = 350$ minutes. (Doing it the other way round gives the same answer: $2 \times 35 = 70$ minutes a day, and $5 \times 70 = 350$ minutes.)

350 minutes

12. 6 groups, each needing a \$3 pack of paper and 2 glue sticks at \$2 each; Ms Park starts with \$50. How much is left, in dollars?

Solution: *Step 1 — what does one group cost?* The glue is $2 \times \$2 = \4 , and with the paper that is $\$3 + \$4 = \$7$ for one group. *Step 2 — what do all six groups cost?* $6 \times \$7 = \42 . *Step 3 — how much is left?* $\$50 - \$42 = \$8$. 8